

6. Project Testing

Testing Approach

Manual testing was performed using FastAPI’s built-in Swagger UI (/docs) for backend endpoints, and browser-based testing for frontend integration.

Test Cases

Test Case	Steps	Expected Result	Status
User Registration	POST /register with valid data	200 OK, user created with ID	Pass
Duplicate Email Registration	Register with existing email	400 error, “email already registered”	Pass
User Login (valid)	POST /login with correct credentials	200 OK, JWT token returned	Pass
User Login (invalid)	POST /login with wrong password	401 error	Pass
Add Loan Record	POST /debt-record with valid data	200 OK, record created with AI strategy	Pass
Get Loans List	GET /loans with valid email	200 OK, array of loans returned	Pass
Delete Loan	DELETE /loans/{id}	200 OK, record removed	Pass
AI Negotiation Strategy	POST /negotiate	200 OK, strategy text returned	Pass
Financial Health	GET /financial-health	200 OK, correct calculations	Pass
CORS Configuration	Frontend calls backend	No CORS errors	Pass

Known Limitations

- The SQLite database resets on backend redeploy (a limitation of the free hosting tier)
- The credit score feature is not implemented, since no real data source is available